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Blog

All blog-users where username & password match given values

```
db.User.find({_id: "alice", password:"password123"});
```

_id	birthday	country	displayName	password
alice	1990-04-15T00:00:00.000Z	USA	Alice Johnson	password123

Query 2: all blog-entries being written by a certain blog-user (username)

```
db.Entry.find({ authorId: "alice" });
```

_id	authorId	categoryId	commentsAllowed	content	creationD	ationDate	description	editDates	impressionCount	title
69d7f427aeef7c42429fec31	alice	69d7f427aeef7c42429fec2e	true	{"text": "This is my first post about technology!", "links": ["http	2026-04-09T	4-09T18:47:03.393Z	An introduction to technology trends.	[]	0	My First Tech Blog
69d7f427aeef7c42429fec35	alice	69d7f427aeef7c42429fec2e	true	{"text": "Let's explore some deep tech insights in AI and blockchai	2026-04-09T	4-09T18:47:03.819Z	Alice shares advanced insights on tech trends.	[]	0	Advanced Tech Thoughts

All blog-entries that does not contain ANY information in one of your additional fields (one of the additionalfields you added to your content).

```
db.Entry.find({"content.links": {$size: 0}});
```

_id	authorId	categoryId	commentsAllowed	content	creationDa	creationDate	description	editDates	impressionCount	title
69d7f911aeef7c42429fec42	bob	69d7f911aeef7c42429fec3f	true	{"text": "Italy is amazing! Here's what I saw...", "links": [], "im	2026-04-09T1	2026-04-09T19:08:01.336Z	Travel diary of my Italy trip.	[]	0	Exploring Italy

all blog-entries where the entry has more than 1 image

```
db.Entry.find({"content.images": {$exists: true, $type: 'array', $ne: [] }});
```

_id	authorId	categoryId	commentsAllowed	content	creationDa	description	editDates	impressionCount	title
69d7f911aeef7c42429fec42	bob	69d7f911aeef7c42429fec3f	true	{"text": "Italy is amazing! Here's what I saw...", "links": [], "im	2026-04-09T1	Travel diary of my Italy trip.	[]	0	Exploring Italy

All blog-entries where the entry contains image(s) and links

```
db.Entry.find({"content.images": {$exists: true, $type: 'array', $ne: [] }, "content.links": {$exists: true, $type: "array", $ne: []}});
```

no output here

All blog-entries whose author either has a given lastname (we don't have lastname so we use displayName) or has the value ‘admin’ but not ‘Guest

```
db.Entry.aggregate(
    [
        {
            $lookup: {
                from: "User",
                localField: "authorId",
                foreignField: "_id",
                as: "author"
            }
        },
        {
            $match: {

                $or: [
                        { "author.displayName": "Alice Johnson" },
                        {$and: [
                            { "author.displayName": {$ne: "Guest"}},
                            {authorId: "admin"}
                        ]}
                    ]
            }
        },
        {
            $project: {
                author: 0
            }
        }
    ]
);
```

	{} _id	{} authorId	{} categoryId	{} commentsAllowed	{} content	{} creationI
1	69d7fa1baeef7c42429fec52	alice	69d7fa1aaeef7c42429fec4f	true	{"text": "This is my first post about technology!", "links": ["http	2026-04-09T
2	69d7fa1baeef7c42429fec56	alice	69d7fa1aaeef7c42429fec4f	true	{"text": "Let's explore some deep tech insights in AI and blockchai	2026-04-09T

{} creationDate	{} description	{} editDates	{} impressionCount	{} title
2026-04-09T19:12:27.021Z	An introduction to technology trends.	[]	0	My First Tech Blog
2026-04-09T19:12:27.467Z	Alice shares advanced insights on tech trends.	[]	0	Advanced Tech Thoughts

all blog-entries where the title is mentioned in the content as well.

```
db.Entry.aggregate([
        {
            $match: {
                $expr: {
                    $regexMatch: {
                        input: "$content.text",
                        regex: "$title"
                    }
                }
            }
        }
    ]
);
```

	{} _id	{} authorId	{} categoryId	{} commentsAllowed	{} content	{} creationDa
1	69d7fd44aeef7c42429feca4	alice	69d7fd43aeef7c42429feca1	true	{"text": "This is my first post about technology!, My First Tech Bl	2026-04-09T1

	{} creationDate	{} description	{} editDates	{} impressionCount	{} title
1	2026-04-09T19:25:56.289Z	An introduction to technology trends	[]	0	My First Tech Blog

all BlogUser sorted ascending by username.

```
db.User.aggregate(
    [
        {
            $sort: {_id: 1}
        }
    ]
);
```

	{} _id	{} birthday	{} country	{} displayName	{} password
1	alice	1990-04-15T00:00:00.000Z	USA	Alice Johnson	password123
2	bob	1985-09-20T00:00:00.000Z	Canada	Bob Smith	securePass
3	carol	1992-07-10T00:00:00.000Z	UK	Carol White	passCarol1
4	dave	1988-02-25T00:00:00.000Z	Australia	Dave Brown	davePass99
5	eve	1995-11-05T00:00:00.000Z	Germany	Eve Black	evePass77

the newest (creationDate) 2 blog-entries.

```
db.Entry.aggregate(
    [
        {
            $sort: {creationDate: -1}
        },
        {
            $limit: 1
        }
    ]
);
```

	_id	authorId	categoryId	commentsAllowed	content	creationDa
1	69d7fd44aeef7c42429feca8	alice	69d7fd43aeef7c42429feca1	true	{"text": "Let's explore some deep tech insights in AI and blockchai	2026-04-09T1

		creationDate	description	editDates	impressionCount	title
2	tech insights in AI and blockchai	2026-04-09T19:25:56.711Z	Alice shares advanced insights on tech trends.	[]	0	Advanced Tech Thoughts

Second oldest blog entry

```
db.Entry.aggregate(
    [
        {
            $sort: {creationDate: 1}
        },
        {
            $skip: 1
        },
        {
            $limit: 1
        }

    ]
);
```

	_id	authorId	categoryId	commentsAllowed	content	creationDa
1	69d7fd44aeef7c42429feca5	bob	69d7fd44aeef7c42429feca2	true	{"text": "Italy is amazing! Here's what I saw...", "links": [], "im	2026-04-09T1

		creationDate	description	editDates	impressionCount	title
1	taly is amazing! Here's what I saw...", "links": [], "im	2026-04-09T19:25:56.392Z	Travel diary of my Italy trip.	[]	0	Exploring Italy

all blog-entries created within the last week containing a link

```
db.Entry.find(
    {
        creationDate: {$gt: new Date(new Date().getTime() - 1000*60*60*24*7)},
        "content.links": {$exists: 1, $type: "array",  $ne: []}
    }
);
```

	_id	authorId	categoryId	commentsAllowed	content	creationDa
1	69d7fd44aeef7c42429feca4	alice	69d7fd43aeef7c42429feca1	true	{"text": "This is my first post about technology!, My First Tech Bl	2026-04-09T1
2	69d7fd44aeef7c42429feca8	alice	69d7fd43aeef7c42429feca1	true	{"text": "Let's explore some deep tech insights in AI and blockchai	2026-04-09T1

		creationDate	description	editDates	impressionCount	title
1	out technology!, My First Tech Bl	2026-04-09T19:25:56.289Z	An introduction to technology trends	[]	0	My First Tech Blog
2	tech insights in AI and blockchai	2026-04-09T19:25:56.711Z	Alice shares advanced insights on tech trends.	[]	0	Advanced Tech Thoughts

the 2 newest blog-comments added to a given username's entries

```
db.User.aggregate(
    [
        {
            $match: {
                _id: "alice"
            }
        },
        {
            $lookup: {
                from: "Entry",
                localField: "_id",
                foreignField: "authorId",
                as: "entries"
            }
        },
        {
            $lookup: {
                from: "Comment",
                localField: "entries._id",
                foreignField: "entryId",
                as: "comments"
            }
        },
        {
            $unwind: "$comments"
        },
        {
            $sort: {
                "comments.creationDate": -1
            }
        },
        {
            $limit: 2
        },
        {
            $project: {
                _id: 0,
                comments: 1
            }
        }
    ]
);
```

```
{} comments
1 {"_id": new ObjectId("69d7fd44aeef7c42429fecae"), "text": "Loved the deep dive into AI!", "likes": new NumberLong("4"), "creationDate": new ISODate("2026-04-09T19:25:56.868Z"),
2 {"_id": new ObjectId("69d7fd44aeef7c42429fecad"), "text": "Great insights, Alice!", "likes": new NumberLong("5"), "creationDate": new ISODate("2026-04-09T19:25:56.867Z"), "entr
```